

Ethan Buckley

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Education

University College London, MSci Natural Sciences

Sep 2024 to Jun 2028

- Physics major, Physical Chemistry minor.
- Relevant modules: Mathematical Methods III, Quantum Physics, Thermal and Statistical Physics.

Experience

Data Science and Research Intern, Down Syndrome Education International

Jun to Sep 2026

- Built hierarchical Bayesian models (PyMC) of vocabulary growth in children with Down syndrome, fitted to 1,424 assessments pooled from 14 datasets in six countries, with Gaussian-process age trajectories and random effects for each study and child.
- Wrote the intention-to-treat, difference-in-differences and mediation models for a Bayesian re-analysis of a randomised reading trial (54 children, four waves) within the study's causal DAG, and ranked predictors with LightGBM and SHAP.
- Built the pipeline that generated the charity's teaching images and audio from a written specification, and the tools that audit, repair and upload teaching assets to its catalogue as drafts. Chose each image background to contrast with its subject, because reduced contrast sensitivity is common in Down syndrome.
- Chose text-to-speech voices by blind A/B listening across four commercial providers, ranked with a Bradley-Terry model, and fixed a scoring bug that gave first place in tied comparisons to whichever voice sorted first.

Literature review, UCL Centre for Computational Science

From Oct 2026

- Will review the literature on hybrid quantum and GPU computing for biomolecular simulation, including variational quantum algorithms. Supervisor: Professor Peter Coveney.

AI Training Specialist (freelance), DataAnnotation

Jun to Aug 2026

- Evaluated large language model answers in physics, chemistry, maths and machine learning, ran any code they contained, wrote reference solutions and produced comparison data.

Seasonal Team Member, Roadchef, Killington Lake

Jul 2025 to Apr 2026

- Cleaned the motorway services site, took in cleaning-supply deliveries for its outlets and covered shifts at Costa, WHSmith and Days Inn.

Work experience, BAE Systems (Barrow) and EDF Energy (Heysham 1)

Jul to Aug 2023

- Spent three days with engineering teams at Barrow and five days with the chemistry and maintenance departments at Heysham 1, including observing reactor-chemistry monitoring.

Projects

- **S&P 500 stock screener (independent):** XGBoost classifier with 22 features plus FinBERT sentiment, validated walk-forward over 1,659 out-of-sample days (2020 to 2026) on a point-in-time universe. Top-15 precision, a classifier metric, was about 46% against a 29% base rate; a volatility-only sort matched it, so the ranking comes from volatility. After trading costs, a portfolio of the top 15 trailed the S&P 500. Some survivorship bias remains.
- **SEIQR epidemic cellular automaton (UCL group project, individually extended):** Vectorised the simulation for a roughly 40-fold speed-up, validated it against the mean-field ODE limit, and made runs reproducible with bit-identical regression tests, and added a Streamlit dashboard.

Skills

- **Programming:** Python (NumPy, pandas, PyMC, XGBoost, LightGBM, SHAP, scikit-learn, pytest), SQL and DuckDB, Git and GitHub Actions, Quarto, LaTeX, generative-model batch APIs.
- **Statistics and machine learning:** Bayesian inference and MCMC, hierarchical models, Gaussian processes, gradient boosting, causal inference (DAGs, intention-to-treat, difference-in-differences, mediation), blind A/B evaluation with Bradley-Terry ranking, NLP with FinBERT.

Activities and interests

- Player, UCL Men's Hockey Club; volunteer junior coach, Kirkby Lonsdale Hockey Club.
- UCL AI Society and Natural Sciences Society.
- Interests: mountaineering, skiing, running.